Manual

Material Monitoring

MPL-MMO 8.1

Version 1.0.662

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Contents

1 Material Monitoring

Overview

Purpose

The Material Monitoring function package extends Material and Production Logistics and/or Tracking & Tracing by functions for monitoring minimum storage periods and/or minimum shelf-lives and consequently contributes to increasing process security.

Implementation considerations

You use the Material Monitoring function package if you apply the Material and Production Logistics and/or Tracking & Tracing function packages, and

- Need to secure compliance with the shelf lives of incoming raw materials;
- Need to secure compliance with minimum shelf lives for semi-finished products and/or WIP material;
- Need to secure minimum storage times for semi-finished products and/or WIP material within production.

Integration

This function package is integrated in the function packages Material and Production Logistics as well as Tracking & Tracing, by:

- extending the Batch object by production and shelf life data;
- verifying the minimum shelf life when incoming batches are registered.

Features

- Configuration of minimum storage period and expiration period
- Monitoring of expiration and warning times
  - Batch and lot-related monitoring of dates and periods
- Warning report
  - Warning report of all lots and batches whose warning date and/or expiration date has been exceeded or will be reached at the evaluation day.
- Expiration statistics
  - Expiration statistics including evaluation as to which material quantities have expired and/or will regularly expire in a given period (table and graph).
- Expiry preview

- Expiry preview including indication as to when which material quantities will have expired, including a forecast of material expiring on basis of the present inventory (table and graph).

2 Configuration of Minimum Storage Times and Expiry Durations

Usage

You use minimum storage time to ensure processing-related waiting times within production and to prevent input materials with a minimum storage time that has not expired from being logged on.

You use warning time to be notified in advance in the evaluations warning report and expiry preview that material is due to expire.

You use expiry limit to define the maximum storage life of a material.

Requirements

You defined material types for the materials you use.

Procedure

Define on which level the times should be maintained.

- You can update the settings on the material type level in the material type configuration.
- When assigning material to material type, you can overwrite the material type settings for selected materials.

Update the times on the level you have selected.

Results

The times are defined.

3 Activating Material Monitoring

Usage

You activate the material monitoring job if you would like the system to automatically implement the status of known batches in the system based on durations defined in the system.

Requirements

You defined the process time defaults for minimum storage time, warning time and/or expiry time in material type or in assignment of material to material type.

Procedure

Make an entry in the Scheduler for the cyclic call of the monitoring job and define the call interval, if it is different.

Parameter name	**Value**
For uploading material receipts:	
Product key	MPL-MMO
License key	MPL-MMO
Command (Windows):	sh.exe ./mpl_lsta.scr
Command (Unix):	./mpl_lsta.scr
Comment:	MPL - Status monitoring
Interval	5

The job is run every five minutes in the system standard delivery. If this interval is sufficient for you, no further adjustments are required.

Results

The system runs the job cyclically at the defined interval.

4 Monitoring Availability Dates

Requirement

You defined the process time defaults for minimum storage time, warning time and/or expiry time in Material type or in Assignment of material to material type.

You scheduled the job in the scheduler.

Procedure the system follows during cyclic monitoring

Batches, which because of the *minimum storage time* configuration, were set to status "M" (minimum storage time) at the time the batch was created are set to status "F" (free) once the minimum storage time has expired.

A batch is considered expired if the period of time between when the batch was created and the current time is greater than the period of time defined in the *Expiry limit* configuration. If this is the case, the batch is set to status "V" (expired).

The status modification as a result of the processing step is logged and is shown in the batch history.

Procedure the system follows during input batch logon

When a batch is logged on as an input batch, a plausibility check is run to verify the minimum storage time and the expiry time.

5 Warning Report

Summary

Menu	Material management→ Inventory management→ Warning report
Transaction code	warnrp
Function authorization	Warnrp

Usage

All batches will be shown in the warning report, for which the warning date and/or expiration date has been exceeded or will be reached at the evaluation day. The data display corresponds mostly to the type of display in batch data maintenance.

Selection criteria

The following selection criteria are available in the application:

Batch

Batch number (entry of wildcards possible)

Status

Batch status (selection of several statuses possible)

Material

Article/item (entry of wildcards possible)

Material type

Material type (entry of wildcards possible)

Material buffer

Material buffer (entry of wildcards possible)

Date, time, relating to

Key date and time related to an expiration date and/or warning time.

Field descriptions

Last modification

- date and time of a batch's last modification
- date of a batch's last modification
- time of a batch's last modification

Modified by

Editor of the last batch's modification

Batch number

Batch number

Quantity

Original quantity

Remaining quantity

Remaining and/or current quantity of the batch

Unit

Quantity unit of the batch

Manufacturing date

Date of creation of a batch

Workplace

Producing workplace of a batch

Batch class

Batch class

Batch information

Information on the batch

Availability date

Date, from which on the batch will be available

Warning date

Warning date of the batch

Warning time

Warning time of the batch

Expiry date

Expiry date of the batch

Status change

Date of the last status change of the batch

Remaining quantity indicator

This indicator shows whether a remaining quantity is assigned to the batch (Y/ N)

Input batch

Input batch number

User

Staff badge number

Batch status

Batch status (free, running, processed, etc.)

Transport status

Transport status of the batch

Workplace

Workplace at which the batch is currently running. Otherwise this is empty.

MES order number

Order number for which the batch is running. Otherwise this is empty.

Reason

Scrap reason

Material

Material number of the batch

Material designation

Designation/ description/ name of the material

Material type

Material type of the batch

Material buffer

Material buffer of the batch

Designation

Designation of the material buffer

Transport unit

Transport unit of the batch

Person

Person, who processed the batch.

Field 1 - 10

Specific batch data (optional)

Production order

Production order, from which the batch was created.

6 Expiry Preview

Summary

Menu	Material management → Inventory management → Expiry overview
Transaction code	eprev
Function authorization	eprev

Usage

The expiry preview deals with this question: "When does how much material expire?" It shows a prognosis of the expired material based on the current stock.

Requirement

All batches that currently have status "F" (Free) or "M" are considered (batch must first be stored for a minimum period before it is available = free). Depending on the selection, consumption is compressed into a specified time frame (daily or hourly).

Selection criteria

The following selection criteria are available in the application:

Material/ material type/ material buffer

Selection of the object to be selected

Material

All materials with the selected material number are selected.

Material type

All data records with the material type are selected.

Material buffer

All data records in the corresponding material buffer are selected.

Grid spacing day/ hour

The expiry preview outputs the selection of the grid spacing in days or hours.

Field descriptions

Index

Consecutive number

Date

Date on which the material expires

Quantity

Available batch quantity

Expired quantity

Quantity that expires on the corresponding date

Expired quantity (accumulated)

Total quantity that has expired by this date

Material buffer

Material buffer that includes the expiring material.

7 Expiry Statistics

Summary

Menu	Material management → Inventory management → Expiry statistics
Transaction code	estat
Function authorization	estat

Usage

The expiry statistics displays how much material has expired within a specific period or how much generally expires within a specific period.

Integration

The display of the expiry statistics refers to the material entered, the material type or material buffer entered and the period indicated with a grid spacing in the specified cycle. Possible grid values are: 5, 15, 30 minutes, 1, 5, 12 hours, daily, weekly, total.

Selection criteria

The application provides the following selection criteria:

Material

Only the expiry statistics for the selected material number is displayed.

Material type

Only the expiry statistics for the selected material type is displayed.

Evaluation mode

The evaluation mode indicates the grid spacing of the expiry statistics.

Material buffer

Only the expiry statistics for the selected material buffer is displayed.

Date from ... to ...

Only the expiry statistics for the selected period is displayed

Field Descriptions

Index

Consecutive number

Evaluation time

Point in time of the material expiration

Material

Material number

Material designation

Material designation

Material type

Material type

Quantity

Quantity

Unit

Quantity unit

Detail applications

Table display

In the table display, all of the fields returned by the data source are displayed. A sum is calculated with regard to the quantity. Grouping and subtotals are possible.

Expired quantity per point in time on the grid

Quantity of material that has expired by the respective grid period. The quantity displayed depends on the selection criteria made and can even contain quantities with differing quantity units.

Expired quantity per material in the evaluation period

Quantity of the respective material that has expired in the specified period.

8 Batch Data Overview

Summary

Menu	Material management → Inventory management → Batch data overview
Transaction code	batov
Function authorization	batov

Usage

The batch data overview is used to display or process one or more batches depending on the chosen selection criteria entered.

Integration

The batch data overview shows all batches included in the data set that were specified by the selection criteria entered.

In the batch data overview, the selected batches can be displayed in their current status together with detail information.

Selection criteria

The application provides the following selection criteria:

"Batch" category

Batch number

The batch number represents the batch ID used by the user. This can be:

- for the input/output at terminal dialogs
- for tracing or information search at the office client

When the configuration option "automatic generation of batch no. when creating batches" is activated, a batch number will still only be assigned automatically as long as the batch number field in the editing dialog box is left empty when a batch is added.

Material

Entering the material number as a selection criterion displays all batches that have this material number.

Workplace

Entering the workplace as a selection criterion displays all batches that have been produced at this workplace/machine.

MES order number

MES order number (order/operation number) by which the batch has been produced.

Internal batch number

The batch number is a unique, system-wide batch identification number.

Material buffer

The individual batches are located in a specific material buffer. With this selection, all batches with the selected material buffer are displayed.

Material type

The individual batches or materials belong to a material type. The same transport and handling guidelines are used for these material types across the system. With this selection, all batches with the selected material type are displayed.

Material category (type)

The material type is used to assign batches to specific classes/ groups. With this selection, all batches with the selected material type are displayed.

Material designation

When the output batch is created on the shop floor stations, the material designation of the currently registered order is used. Because no material master is managed, the material designation is saved redundantly in the batch description.

Manufacturing date from / until

Date/period of the production of a batch

Consider long-term data

Archived data may be selected.

"Status" category

Batch class

The batch class describes the overall quality of the batch. With this selection, all batches matching the selected batch class are displayed.

Batch status

The batch status describes the technical system and production status of a batch. Selecting the batch status as a selection criterion displays all batches that have this status.

Quality status

The quality status "blocked" prevents a batch from being logged on. Selecting the quality status as a selection criterion displays all batches that have this status.

Manual Q status

Selecting the manual Q status as a selection criterion displays all batches that have this status.

Material status

The material status indicates a logical status of the batch, e.g. packed, tested. Selecting the material status as a selection criterion displays all batches that have this status.

Transport status

The transport status represents the technical system status with respect to transfer postings to external storage. Selecting the transport status as a selection criterion displays all batches that have this status.

Advance logon flag

Restriction to batches logged on in advance.

"Attributes" category

Attribute (1 to 10)

The data displayed may be restricted to the batch attributes directly kept for the batch.

"Batch attributes" category

Batch attribute (designation)

The data displayed may be restricted to the batch attributes configured for the material type. There are 40 text fields, 20 numeric fields and 20 decimal fields that may be configured altogether.

"Alternative batch numbers" category

Alternative batch number (1 to 20)

Selecting an alternative batch number as a selection criterion displays all batches that currently have this identifier.

"Reservation" category

Reserved for order

Entering or selecting the order number in the field displays all batches that were produced for this order/ OP.

Reserved for OP

Entering or selecting the order/OP number in the field displays all batches that were produced for this order/ OP.

"Dates" category

Expiry date from / until

Date/period that indicates the shelf life of a batch

Availability date from / until

Date/period of the availability of a batch

Warning time

Warning date of a batch

"Miscellaneous" category

Serial number

Selecting a serial number as a selection criterion displays all batches for which this serial number is entered in the serial number field.

Batch (LOT) number

Selecting the batch/lot number as a selection criterion displays all batches that currently have this identifier.

Person

Selecting the person as a selection criterion displays all batches that have been produced by this person and, as a result, are currently assigned to this person (identifier).

Merged batch

Selecting the merged batch number as a selection criterion displays all batches for which the merged batch number is entered in the merged batch field.

PPS batch

Selecting the PPS batch as a selection criterion displays all batches for which the PPS batch number is entered in the PPS batch field.

Editing functions

These functions are provided by the standard features for creating, editing, etc. to edit one or several data records:

 **Add batch**

This function can be used to insert a new batch. Goods movements are not generated for the new batch.

 **Copy batch**

This function can be used to copy a batch. However, the batch number for the new batch has to be entered anew. Goods movements are not generated for the new batch.

 **Edit batch**

A batch can be edited using this function. But neither cancellations nor goods movements are generated for the changed batch.

 **Delete batch**

This function can be used to delete a batch. But neither cancellations nor goods movements are generated for the deleted batch.

Pool batches (new batch number)

Pooling/merging batches. A new batch number is generated for the new batch (merging batches).

When pooling merged batches, all individual batches are assigned to a new merged batch number.

Pool batches (use existing batch numbers)

Pooling/merging batches. A batch number already included in the pool/merge is used for the new batch (merging batches).

When pooling merged batches, all individual batches are assigned to the selected merged batch number that is part of the pool.

Split batch

A new batch is created by splitting it off from an existing batch (splitting batches). In this case, the following options exist for the remaining target quantity of the original batch:

- Repost remaining quantity of batch to new batch
- Reduce existing batch to the remaining quantity

If a merged batch is split, the selected individual batches/serial numbers are transferred to a split off batch. The remaining individual batches/serial numbers remain assigned to the original merged batch.

Repost

This function can be used to repost a batch to another material buffer (Reposting batches).

Generate

The function "enter goods receipt batch" is used to create batches manually (Generating batches). This is necessary if material is delivered via the goods receipt, for example.

Edit batch attribute

The batch attributes of the batch selected in the grid are shown in the tab “batch attributes”. Only those batch attributes will be shown that have been assigned the indicator “show attribute on client” within the configuration of batch attributes.

Provided that batch attributes are shown, they can be edited by clicking the button “edit batch attribute”. The field types/field lengths specified for the configuration of batch attributes are not checked.

Go to

Graphic batch tracing

Starts the application Graphic batch tracing.

Document management

Starts the document management

"Batch data overview" detail application

In the batch data overview detail application, all batches are displayed according to the selections entered.

The information or field descriptions of the selected batches and, thus, of the detail application are described in the documents entitled batch object and batch structure.

9 Requirements Overview

Summary

Menu	Material management → Inventory management → Requirements overview
Transaction code	reqov
Function authorization	reqov

Usage

The function “material requirements” provides an overview of the materials required at machines. It is particular to this function that material requirements are exactly determined by day taking the shift calendar into account. Consequently, material requirements are distributed according to the run time of OP(s). “By day” means here “by shift”, i.e. the day starts with the beginning of shift 1 and ends with the end of the last shift (e.g. shift 3).

Requirements

The PPS system transfers the materials required for the production of an operation and, as a result, they are available in the component list of the OP within the system.

Please note:

The determination of material requirements depends, among others, on the planned start date of an operation. For this reason, the planned start date of an operation needs to be transferred correctly by the PPS system or planned in shop floor scheduling. If, however, the order sequencing function is used this planned date is used to define the order of processing (internal algorithm). In this case, the planned start date can no longer be derived.

Selection criteria

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The application provides the following selection criteria:

Material

Shows all materials assigned to the selected material.

Workplace

Shows all materials assigned to the selected workplace.

MES order number

Shows all materials assigned to the selected MES order number.

Material type

This selection criterion refers to the material type of the material. All materials assigned to the selected material type are shown.

Planned start

Shows all materials assigned to the selected planned start.

Consider pool of groups

Defines whether or not the pool of groups is to be taken into account for selection.

10 Stock Overview

Summary

Menu	Material management→ Inventory management→ Stock overview
Transaction code	stov
Function authorization	stov

Usage

The stock overview is an evaluation of the material management. Here, the employees of the warehouse, dispatch and production supply or logistics departments can inspect the current buffer stock and/or stock in production.

Integration

This evaluation shows the batch and material stock within a material buffer cumulated for each material and each batch status within the material, i.e. the material number. Such batches with a quantity of 0 and batches that are not assigned to a material buffer will not be taken into account.

Selection criteria

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The following selection criteria are available in the application:

General

Batch number

(Externally used) batch number that is to be selected.

Material

All materials with the selected material number will be shown.

Internal batch number

Internal batch number that is unique within the system and that is to be selected.

Material buffer

This selection criterion refers to the material buffer of the batch. All batches with the selected material buffer will be used for the evaluation.

Material type

This selection criterion refers to the material type of the material. All materials with the selected material type will be shown.

Batch status

This selection criterion refers to the batch status of the material. All materials with the selected batch status will be shown.

Attributes

Attributes 1 - 10

Selecting an attribute as a selection criterion displays all inventories of batches that currently have this identifier.

Batch attributes

Batch attributes

Selecting a batch attribute as a selection criterion displays all inventories of batches for the material type of which this batch attribute has been configured and that are currently assigned this identifier.

Alternative batch numbers

Alternative batch number 1-20

Selecting an alternative batch number as a selection criterion displays all inventories of batches that currently have this identifier.

Field description

Primarily, the evaluation shows the current information on a batch as well as information on the material buffer concerned.

Information on batch

The information per batch is the same as the one presented in the batch data overview.

Information on material buffer

In the stock overview the information on the material buffer will be shown per batch. This is primarily:

Material buffer

Material buffer of the batch

Designation

Designation of the material buffer

Storage location

Assigned storage location of the material buffer

Type

Type of the material buffer

11 TPU Stock Overview

Summary

Menu	Material management→ Inventory management→ TPU stock overview
Transaction code	stovhu
Function authorization	stovhu

Usage

The transport units stock overview is used to show the quantities of materials in the individual transport units. This can be used to determine which quantities of which material and which batches are included in which transport unit and how the transport units are currently assigned.

Integration

A pre-selection of material, transport unit and of other criteria will be shown as cumulative display of the batch/ material stock per transport unit. The display will be accumulated for each material, transport unit and batch status. Batches with a remaining quantity of 0 will not be accounted for.

Only those materials will be accounted for that are assigned to those material types, for which the "inventory management" indicator is not set to No.
Moreover, only those transport units will be accounted for, for which the "inventory management" indicator is set to "Y".

Selection parameters

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The following selection criteria are available in the application:

General

Batch number

All materials with the selected batch number will be shown.

Material

All materials with the selected material will be shown.

Material type

This selection criterion refers to the material type of the material. All materials with the selected material type will be shown.

Batch status

This selection criterion refers to the batch status of the material. All materials with the selected batch status will be shown.

Transport unit

All materials with the selected transport unit will be shown.

Attributes

Attributes 1 - 10

Selecting an attribute as a selection criterion displays all inventories of batches that currently have this identifier.

Batch attributes

Batch attributes

Selecting a batch attribute as a selection criterion displays all inventories of batches for the material type of which this batch attribute has been configured and that are currently assigned this identifier.

Alternative batch numbers

Alternative batch number 1-20

Selecting an alternative batch number as a selection criterion displays all inventories of batches that currently have this identifier.

Field description

Calculation of the values in the "inventory" category:

Number of batches

Quantity of batches (in line with the stock indicator set to the material type)

Quantity

Sum of yield + scrap

Unit

Unit from the configuration table TPU – material type

Yield

Quantity identified as yield that is in fact included in the "number of batches"

Scrap

Quantity identified as scrap that is in fact included in the "Number of batches"

Calc. Quantity

Quantity of material that would fit into the currently assigned transport units:

Quantity from the configuration TPU material type for this TPU x number of batches

Calc. Quantity of TPU

Quantity of TPU that would be necessary for the total quantity:

Quantity/ (quantity from the configuration TPU material type for this TPU)

The result will be rounded up to the next integer.

Batches that are not assigned to a transport unit will **not** be accounted for in this evaluation.

12 Applications provided in MOC:

Application	Documentation	Transaction code	Function authorization
Warning report	MOC_WarningReport.pdf	warnrp	warnrp
Expiry preview	MOC_ExpiryPreview.pdf	eprev	eprev
Scrap statistic	MOC_ExpiryStatistics.pdf	estat	estat
Batch data overview	MOC_BatchOverview.pdf	batov	batov
Requirements overview	MOC_RequirementsOverview.pdf	reqov	reqov
Stock overview	MOC_StockOverview.pdf	stov	stov
Stock overview HU	MOC_StockOverviewHU.pdf	stovhu	stov